MILITARY SPECIFICATION

CARTRIDGE, 9MM, TEST, HIGH PRESSURE, XM905

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Cartridge, 9MM, Test, High Pressure XM905 intended for use in proofing of 9MM weapons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Table for Inspections by Attributes
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition Through Caliber .50

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:35Z
Check the source to verify that this is the current version before use.

MIL-STD-644	-	Visual Inspection Standards and Inspection Procedurs for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
DOD-STD-1468	-	Small Caliber Ammunition Test Procedures, 9MM Cartridges

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US Army Armament Research and Development Center

9378265	-	Cartridge, 9MM, Test, High Pressure, XM905
7643674	-	Classification of Cartridge Case Defects for Small Arms Ammunition
9378266	-	Packaging and Marking, Cartridges, 9MM, Test , High Pressure, XM905, Cartons, Box, Ammunition, M2A1, Box, Wirebound
L19378265	-	Index of Inspection Equipment Lists

Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article inspection. When specified, a sample shall be subjected to first article inspection. (See 4.3) Unless otherwise specified, the first article shall include the pilot pack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

3.2 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.3. Parts. The parts shall comply with all requirements specified on Dwg. 9378265, all associated drawings, and with all requirements specified in applicable specifications.

3.4 Bullet extraction. The average force necessary to extract the bullet from the cartridge case shall be not less than 80 lbs, nor shall any individual extraction force be less than 60 lbs.

3.5 Residual stress. The cartridge case shall not split when subjected to one percent mercurous nitrate solution for 15 minutes.

3.6 Chamber pressure. The corrected average peak chamber pressure of the cartridge at the case mouth position shall be 276 MPa plus or minus 15 MPa when conditioned and fired at 21°C. The standard deviation of the chamber pressures shall not exceed 11 MPa. When conditioned and fired at the following temperatures, the uncorrected average peak chamber pressure at each temperature shall not vary by more than plus or minus 20 MPa from the uncorrected average pressure obtained at 21°C.

0°C

+38°C

3.7 Waterproof. When placed under water and subjected to a one half atmosphere negative pressure for 30 seconds, the cartridge shall not release more than one bubble of air.

3.8 Primer sensitivity. The energy imparted by a 55 ± 0.5 gram steel ball falling 12.0 inches shall initiate all primers. The energy imparted by a 55 ± 0.5 gram steel ball falling three inches shall not cause initiation of the primer.

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.9.2 Foreign matter. The cartridge shall be free of corrosion, stains, discoloration, dirt, oil, and smears of lacquer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3)

b. Quality Conformance Inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The sample shall be manufactured using the same methods and processes proposed for use in manufacturing the production cartridges. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Cartridge, 9MM, Test, High Pressure, XM905	9378265	2,000

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-70509(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Cartridge, 9MM, Test, High Pressure, XM905				9378265
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	*		3.3	4.4.2.1
	Bullet extraction	20		3.4	4.5.2
	Residual stress	50		3.5	4.5.3
	Chamber pressure	90		3.6	4.5.4
	Waterproof	20		3.7	4.5.5
	Primer sensitivity	**		3.8	4.5.1

NOTES:
* All first article sample units shall be subjected to Examination for Defects prior to test.
** See Table II

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4. 4.1 Quality conformance inspection. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Primer from one lot and one manufacturer.

b. Propellant from one lot and one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Specified AQL's for major and minor defects shall be applied on a class basis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9378265
4.4.2.1	Cartridge, 9MM, Test, High Pressure, XM905				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1.	Split case in K, L, or M location (See Dwg. 7643674)		100%	3.3	Visual
2.	Perforated Case		100%	3.3	Visual
3.	No primer		100%	3.3	Visual
4.	Cocked primer		100%	3.3	Visual
5.	Inverted primer		100%	3.3	Visual
MAJOR:			0.25%		
101.	Cartridge, corroded or stained, if etched			3.3	Visual
102.	Case, round head			3.3	Visual
103.	Split case in I or J location (See Dwg. 7643674)			3.3	Visual
104.	Case, beveled underside of head			3.3	Visual
105.	No radius on head (rim)			3.3	Visual
106.	Split bullet jacket			3.3	Visual
107.	Loose bullet			3.3	Visual
108.	Loose primer			3.3	Visual
109.	Total length of cartridge, incorrect			3.3	Gage
110.	Cartridge profile failure (requiring more than 20 lbs. dead weight to insert in profile & alignment gage)			3.3	Gage

NOTES: Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T03:35Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.1	Cartridge, 9MM, Test, High Pressure, XM905				9378265
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111.	Diameter of extractor groove, incorrect			3.3	Gage
112.	Diameter of head, incorrect			3.3	Gage
113.	Thickness of head, incorrect			3.3	Gage
114.	Depth of primer, incorrect			3.3	Gage
115.	Case length, incorrect			3.3	Gage
MINOR:			1.5%		
201.	Cartridge, discolored, dirty, oily, smeared			3.3	Visual
202.	Case dent			3.3	Visual
203.	Case draw scratch			3.3	Visual
204.	Case scratch			3.3	Visual
205.	Case scaly metal			3.3	Visual
206.	Case fold			3.3	Visual
207.	Case wrinkle			3.3	Visual
208.	Case buckle			3.3	Visual
209.	Case bulge			3.3	Visual
210.	Illegible or missing head stamp			3.3	Visual
211.	Defective head			3.3	Visual
212.	Defective mouth			3.3	Visual
213.	Bullet dent			3.3	Visual
214.	Bullet scratch			3.3	Visual
215.	Scaly metal (bullet)			3.3	Visual
216.	Upset (crooked) point (bullet)			3.3	Visual

NOTES: Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T03:35Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70509(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9378265
4.4.2.1	Cartridge, 9MM, Test, High Pressure, XM905				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
217.	Flat point (bullet)			3.3	Visual
218.	Nicked or dented primer			3.3	Visual
219.	No waterproofing material (primer pocket joint)			3.3	Visual
220.	Workmanship			3.12	Visual

NOTES: Refer to MIL-STD-636 for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

4.4.3 Tests. The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.5.

4.4.3.1 Test samples. The quantities for the various tests shall be as specified in Table II. The test samples shall be selected from lots that have met the visual and dimensional inspection requirements.

TABLE II. Tests.

TEST	Number of Cartridge 0°C	Ambient	+38°C
Primer sensitivity 1/		1/	
Bullet extraction 2/		20	
Residual stress 2/		50	
Chamber pressure 2/ 3/	30	30	30
Waterproof 2/		20	

1/ This test shall be performed prior to all other tests cited in Table II. Test sample shall consist of number of cartridges required to perform a complete run-down test wherein 50 primers are tested at each height. If the average critical height (H) plus five standard deviations (5 sigma) exceeds the all fire requirement or if the average critical height minus two standard deviations (2 sigma) is less than the no fire requirement, the lot shall be rejected, subject to performance of a second complete run-down test wherein 100 primers are tested at each height. Failure of the second sample to comply with the above criteria shall be cause for rejection of the lot.

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot, subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ Firing defects shall not exceed the acceptance number specified in Table III.

TABLE III. Firing defect.

DEFECTS	ACCEPTANCE NUMBER	
	First Sample	Cumulative (Combined First and Second Sample)
Misfire	0	1
Bullet remaining in bore (1)	0	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

TABLE III (Continued)

Blown primer (2)	0	1
Loose primer which falls out of pocket	0	1
Case split (longitudinal)		
(i) Mouth (defined as the length of bullet intrusion)	2	5
(ii) Body	1	2
(iii) Through the head	0	1
Case rupture (circumferential)		
(i) Partial rupture	0	1
(ii) Complete rupture	0	1

(1) No re-test permitted. Lot shall be rejected.

(2) Primer not in pocket after firing and both head of cartridge case and pocket enlarged and distorted.

4.4.4 Packaging, packing, and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of any incorrect type other than high pressure test shall be classed as a major defect. All non-conforming cartridges shall be rejected.

4.4.5 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the Paragraph Reference/Inspection method column in the table starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.4.5.1 Index of inspection equipment list. The examination tests shall be made using equipment listed on L19378265.

4.5 Test methods and procedures.

4.5.1 Primer sensitivity. The test shall be conducted in accordance with DOD-STD-1468.

4.5.2 Bullet extraction. The test shall be conducted in accordance with DOD-STD-1468.

4.5.3 Residual stress (mercurous nitrate). The test shall be conducted in accordance with DOD-STD-1468.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

4.5.4 Chamber pressure. The test shall be conducted in accordance with DOD-STD-1468.

4.5.5 Waterproof. The test shall be conducted in accordance with DOD-STD-1468 Type II Test.

4.5.6 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

5. PREPARATION FOR DELIVERY

5.1 Packing. Level A (World Wide Shipment). The cartridges shall be packed in accordance with Drawing 9378266.

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

6. NOTES

6.1 Ordering data. Procurement documents shall be in accordance with MIL-A-48078 and shall include the following:

a. Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purpose .

b. Provisions for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.2 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, AMCCOM, Dover, NJ 07801-5001, ATTN: AMSMC-QAF-I(D).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is no under the cognizance of ARDC.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project No 1305-AB27)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:35Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use i submitting comments and suggestions for improvements. All users of military standardization documents are invited to provic suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, an mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, w too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTA
NECESSAI
IF MAILE
IN THE
UNITED STA

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT MUNITIONS & CHEMICAL COMMAND
ATTN: DRSMC-QA
DOVER, NJ 07801 5001

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:35Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70509	CARTRIDGE, 9MM, TEST, HIGH PRESSURE, XM905

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:35Z
Check the source to verify that this is the currentversion before use.